

2(a)	type of treatment (solution) ;	1
2(b)	<i>any two from:</i> 1 volume of, treatment / EchAMP / bacitracin, solution ; 2 concentration of, EchAMP / bacitracin, solution ; 3 concentration / volume , of fluorescence (chemical) ;	2
2(c)(i)	(to) <u>compare</u> (the effect of EchAMP and) <u>no EchAMP</u> on, bacterial / E. coli (population), <u>growth</u> ;	1
2(c)(ii)	(to) compare (the effect of EchAMP to) bacitracin / antibiotic (on bacterial / E. coli (population), growth) ;	1
2(d)(i)	1 axis labels with units on correct axes <u>and</u> completed key ; 2 three correct data points plotted at 7 hours, with data points joined to the correct values at 6 hours ;	2

Question	Answer	Marks
2(d)(ii)	any two from: 1 bacitracin reduces (population growth of <i>E. coli</i>) more than EchAMP (after 6 hours / at 7 hours) ; ora 2 no difference (in effect on <i>E. coli</i>) between 0 and 6 hours ; 3 bacitracin <u>and</u> EchAMP reduce (population growth of <i>E. coli</i>) more than the negative control, after 3 hours / at 6 hours / at 7 hours ;	2
2(e)(i)	there is no difference between the (mean) fluorescence (emitted by <i>E. coli</i> after seven hours when exposed to), the negative control and EchAMP ;	1
2(e)(ii)	1 correct working ; 2 ($t =$) 9.37 when rounded to 3 significant figures ; 3 ($t =$) 9.370 ; must be 4 significant figures	3
2(e)(iii)	any four from: 1 <i>idea that</i> experiment is, not carried out in a person / carried out in a laboratory ; 2 <i>idea that</i> experiment, only used <i>E. coli</i> / did not use other bacteria ; 3 EchAMP may kill beneficial bacteria (in the digestive system) ; 4 EchAMP might not be as effective as other, treatments / antibiotics ; 5 EchAMP could, cause side effects / be toxic / AW, in humans or experiment only carried out for 7 hours / no information about longer term effects ; 6 EchAMP is a protein <u>and</u> would be digested (by enzymes in the digestive system) ; 7 <i>idea that</i> the EchAMP did not kill all of the <i>E. coli</i> ;	4